

Instancia	Movimentos (Manhattan)	Nós (Manhattan)	Tempo (Manhattan) ms
Instancia 01	57	4775600	8892.75
Instancia 02	55	1740739	2967.37
Instancia 03	59	64054952	116930
Instancia 04	56	11639992	19391.6
Instancia 05	56	1342203	2269.03
Instancia 06	52	2251246	4078.54
Instancia 07	52	47107139	87372.8
Instancia 08	50	2676951	4703.35
Instancia 09	46	44219	894.076
Instancia 10	59	25506575	42594.3
Instancia 11	57	12842425	21534.3
Instancia 12	45	101872	181.759
Instancia 13	46	888234	1827.27
Instancia 14	59	26410500	42416.6
Instancia 15	62	57621181	88925.4
Instancia 16	42	1219425	2569.11
Instancia 17	66	176913345	265895
Instancia 18	55	3875782	6623.12
Instancia 19	46	232341	465.346
Instancia 20	52	2182098	3935.43
Instancia 21	54	6330901	10949.6
Instancia 22	59	40913111	71898.1
Instancia 23	49	275786	556.406
Instancia 24	54	9573271	18368.5
Instancia 25	52	5365399	10152
Instancia 26	58	30171809	51524.6
Instancia 27	53	12938949	23893.2
Instancia 28	52	914702	1586.8
Instancia 29	54	2722755	4630.85
Instancia 30	47	342904	573.939
Instancia 31	50	314699	510.382
Instancia 32	59	83012191	141674
Instancia 33	60	10222636	15926.8
Instancia 34	52	696588	1188.75
Instancia 35	55	6552760	10936.8
Instancia 36	52	3977078	6897.51
Instancia 37	58	11968465	19326
Instancia 38	53	2372596	3799.58
Instancia 39	49	1717230	3275.92
Instancia 40	54	6817079	12189.6
Instancia 41	54	8220637	15733.8
Instancia 42	42	135181	253.068
Instancia 43	64	5517506	7583.17
Instancia 44	50	1295346	2438.21
Instancia 45	51	806143	1362.9
Instancia 46	49	729611	1259.76
Instancia 47	47	117885	218.25
Instancia 48	49	125167	220.646
Instancia 49	59	183002699	321492

Movimentos (PDB)	Nós (PDB)	Tempo (PDB) ms	
57	3956177	42156.1	<- 1 exec
55	1231025	2861.53	
59	46099203	117508	
56	7336050	16482.4	
56	679340	1565.12	
52	1452316	3631.44	
52	19310365	50596.5	
50	1733463	4175.23	
46	25545	65.338	
59	17891335	41609.2	
57	8284233	19549.4	
45	57751	158.875	
46	298911	845.471	
59	17262261	39660.1	
62	39112955	86731.3	
42	653673	1845.99	
66	71150720	154061	
55	2749804	6496.41	
46	166324	473.424	
52	1540065	3972.38	
54	5157000	12951.3	
59	21848418	59432.8	
49	245018	775.661	
54	7201887	19829.5	
52	4012818	10671.6	
58	21139979	51552.6	
53	9410454	23901	
52	479132	1240.69	
54	1807455	4364.93	
47	214911	519.657	
50	220273	514.724	
59	49522287	117707	
60	4042713	9330.2	
52	368764	872.488	
55	3671632	8513.53	
52	2050740	5047.99	
58	9176486	21715.6	
53	1757106	3924.51	
49	1222532	3288.88	
54	3502214	8734.29	
54	6050441	16020.3	
42	51862	132.134	
64	3648057	7283.5	
50	986445	2538.14	
51	659626	1574.41	
49	489925	1205.41	
47	62612	164.747	
49	96529	229.023	
59	72364947	181605	

Instancia 50	53	1403173	2382.14
Instancia 51	56	3379076	5603
Instancia 52	56	37644603	65314.6
Instancia 53	64	68173950	112728
Instancia 54	56	8581575	14734.8
Instancia 55	41	42866	907.663
Instancia 56	55	85585045	166024
Instancia 57	50	1540535	2731.48
Instancia 58	51	884707	1659.96
Instancia 59	57	59585617	115391
Instancia 60	66	144353921	217526
Instancia 61	45	374013	707.199
Instancia 62	57	13230180	23079.3
Instancia 63	56	14771633	25729.8
Instancia 64	51	18275308	38874.4
Instancia 65	47	1129288	2185.4
Instancia 66	61	209606405	354243
Instancia 67	50	21955301	44529.8
Instancia 68	51	3903740	7338.94
Instancia 69	53	3824104	6990.97
Instancia 70	52	18697052	35638.1
Instancia 71	44	117397	228.747
Instancia 72	56	45826439	84208.7
Instancia 73	49	569438	1014.04
Instancia 74	56	153625	233.173
Instancia 75	48	3261921	6565.36
Instancia 76	57	9586248	17991.8
Instancia 77	54	3119015	5373.58
Instancia 78	53	1249349	2097.55
Instancia 79	42	50244	981.313
Instancia 80	57	4567040	7594.5
Instancia 81	53	1216567	2075.04
Instancia 82	62	97142617	165437
Instancia 83	49	1263876	2437.12
Instancia 84	55	18044682	31371.4
Instancia 85	44	284817	555.565
Instancia 86	45	128225	243.029
Instancia 87	52	4036834	6899.54
Instancia 88	65	234809966	377880
Instancia 89	54	8934014	17738.6
Instancia 90	50	427694	716.768
Instancia 91	57	58867885	104361
Instancia 92	57	34700004	60907.1
Instancia 93	46	184738	359.247
Instancia 94	53	34420	531.288
Instancia 95	50	455832	819.346
Instancia 96	49	582742	1048.08
Instancia 97	44	148234	314.492
Instancia 98	54	64760468	128587
Instancia 99	57	12060173	20513.3

53	814909	1920.67	
56	1331678	3035.38	
56	27927793	67611.3	
64	56889908	135052	
56	6638737	15123.2	
41	38385	105.766	
55	17310682	43999.5	
50	313919	775.789	
51	175776	431.419	
57	53090867	139172	
66	73527061	150687	
45	230368	577.223	
57	7999334	18579.2	
56	14088514	32405.9	
51	10007633	26988.5	
47	365186	921.098	
61	88832782	192879	
50	9433780	24615.8	
51	2523736	6224.88	
53	2369465	5990.41	
52	6949899	17080.1	
44	90318	224.302	
56	34179152	82159.7	
49	212781	486.529	
56	84708	169.672	
48	1565578	4107.37	
57	6363562	15560.1	
54	986083	2226.36	
53	911903	2019.18	
42	46273	118.701	
57	2987692	6967.87	
53	426185	1067.23	
62	81405889	185747	
49	981867	2507.77	
55	14796457	34588.3	
44	215539	536.09	
45	86171	214.936	
52	2986100	6875.68	
65	164932164	354967	
54	5673518	14933.8	
50	326213	737.225	
57	39992662	94613.8	
57	13681385	30960.5	
46	158909	417.791	
53	28874	626.677	
50	320968	807.171	
49	355791	872.23	
44	135575	371.109	
54	44711293	113087	
57	5457136	11653.4	

Instancia 100	54	17574779	33414.1	
1° Exec PDB 15 puzle	Movimentos	Nos	Tempo (segundos)	
MÉDIA Manhattan	53,05	2,22E+07	3808,72	
MÉDIA PDB	53,05	1,28E+07	3036,08	
MELHORIA	0,00%	42,26%	20,29%	
2° Exec PDB 15 puzle	Movimentos	Nos	Tempo (segundos)	
MÉDIA Manhattan	53,05	2,22E+07	3808,72	
MÉDIA PDB	53,05	1,28E+07	2914,94	
MELHORIA	0,00%	42,26%	23,47%	
		resultados varia por exec		

54	5275357	12755.3	
1° Exec PDB 8 puzle	Movimentos	Nos	Tempo (segundos)
MÉDIA Manhattan	22,16	1123,74	0,120896
MÉDIA PDB	22,16	22,16	0,140298
MELHORIA	0,00%	98,03%	-16,05%
2° Exec PDB 8 puzle	Movimentos	Nos	Tempo (segundos)
MÉDIA Manhattan	22,16	1123,74	0,120896
MÉDIA PDB	22,16	22,16	0,0025906
MELHORIA	0,00%	98,03%	97,86%